



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

What is a data dictionary?



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- What do you do with a messy database?
- How do you manage and organize big amounts of data?

Can Do Statements:

- I can describe important facts related to data dictionary.
- I can explain the concept of data dictionary.
- I can identify key features about data dictionary.





Let's Explore!

Jaime and George discuss about when to use a data dictionary.



Instructions: Read the text below.

- Jaime:** Hi George! My boss told me to talk to you. He said you can help us solve a problem we have in our department. Are you doing anything this afternoon?
- George:** Actually, I am going to the dentist after lunch. I have an appointment, but we can talk right now. Is it OK?
- Jaime:** Yeah, no problem. I want you to give me some advice on how to manage and organize our databases. The supervisor is visiting us next week and everything is a mess.
- George:** OK. I need you to listen to me very carefully. In my opinion, what your department needs is a data dictionary; it is basically a modern database that keeps track of the elements of another database.
- Jaime:** That sounds very interesting. What else can you tell me about it?
- George:** Well, a data dictionary creates a collection of descriptions of separate data in a database. To start with, you have to identify each element and its relationship to other elements; experts call this process "data modeling". It produces a representation of object relationships where all elements receive a name, then the system defines their relationships in a brief text. Finally, it organizes this collection of definitions for reference into a book, this book is the data dictionary.
- Jaime:** A data dictionary seems like the solution we need. Is there anything else you can advise us to do?
- George:** If you develop software that uses a database, a data dictionary is one of the best ways to make quick views, as well as to understand where a data item fits in the structure and what values it can have. In addition, you can also use the hashing function; it is helpful to have items retrieved faster, by using a shorter version of such item.
- Jaime:** Thanks, George! You're very kind.

**Let's Practice!**

What is happening next week?

Instructions: Use the words from the box to correctly complete the sentences.

is presenting a / organizing data / are creating /
keeping track / supervisor is

1. Jaime is _____ into different categories this week.
2. Jaime _____ data dictionary the next week.
3. The _____ checking separate data tomorrow.
4. The software engineers _____ the data dictionary this Tuesday.
5. They are _____ of databases on Wednesday morning.

**Let's Practice!**

The supervisor is giving instructions as to* what needs to be done.

*"As to" has the similar meaning of "about".

Instructions: Match column A with column B. Read the sentences aloud. Check your pronunciation with your instructor.

COLUMN A

1. Do you want me to help
2. I want her to keep
3. I need you to select new items
4. I will ask her to make a clear

COLUMN B

- a. in order to organize them in the structure.
- b. relationship between items in the databases.
- c. track of the databases.
- d. you create the data dictionary?





Extension Activity

Let's find out more about data dictionary!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about data dictionary.
- Summarize the main aspects about it.
- Write a list of its main benefits and organize them in order of importance.
- Prepare a short presentation with all of this information.
- Share your presentation with your partners and teacher.
- Answer the possible questions your partners and teacher may have.

Self-Assessment

- I look for more information about data dictionary.
- I summarize the main aspects about data dictionary.
- I list the most important benefits of data dictionary.
- I organize a presentation with the information I found.
- I talk about data dictionary in front of my class.
- I answer my partners and teacher's questions.

Yes

No

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **appointment** (noun): a meeting; a date
e.g. I have an appointment, but we can talk right now.
- **keep track** (idiom): to follow closely
e.g. A data dictionary is basically a modern database that keeps track of the elements of another database.
- **brief** (adjective): short, concise
e.g. It produces a representation of object relationships where all elements receive a name, then the system defines their relationships in a brief text.
- **organize** (verb): to put in order
e.g. I want you to give me some advice on how to manage and organize our databases.
- **supervisor** (noun): person in charge of reviewing your work
e.g. The supervisor is visiting us next week and everything is a mess.
- **collection** (noun): group of objects accumulated
e.g. The system organizes this collection of definitions for reference into a book, this book is the data dictionary.



Learning Points

Present Continuous for Future Arrangements in Affirmative

We generally use the **present continuous tense** to talk about activities that are developing right now. However, it has another important use, you can also use it to express activities in the future.

Depending on the context, you **use the present continuous tense** to express arrangements or plans. This use implies that more people, apart from the speaker, are conscious and know about the event. It is also implicit that there is certain preparation or planning for the activity to take place or happen.

The structure for the present continuous for arrangements is:

Subject + **auxiliary verb to be** + **main verb (-ing form)** + time complement

Examples:

- *I **am starting** a Java class next Monday.*
- *We **are launching** the program tomorrow morning.*

In the previous examples, you can see that the time complement suggests the action is taking place in the future. Always pay attention to the time complement in the sentence to avoid mistakes, as you can confuse it with the present continuous for actions happening right now.

However, it isn't mandatory to use a time complement each time you need to talk about arrangements using the present continuous tense.

Examples

- *The supervisor is visiting us next week and everything is a mess.*
- *Are you doing anything this afternoon?*

Verbs + Objects + Infinitives.

Certain **verbs** can have a direct **object** followed by another verb in **infinitive**.

There are some verbs that can include a second person in the action, apart from the principal subject or performer of the activity. This secondary character in the sentence is the object of the verb. Some of these verbs include ask, help, invite, need, tell, advice, order, persuade, want, etc.

For example: Janet **asked Luisa to check** the program for bugs.





Learning Points

Remember that we can use pronouns instead of names, so we have subject pronouns as well as object pronouns. Look at the chart:

Subject Pronoun	Object Pronoun
I	Me
You	You
He	Him
She	Her
It	It
We	Us
They	Them

For example:

- **They** told **her** to leave the copy on the desk.
- **He** needs **it** to work before leaving.
- **I** want **you** to clean this mess.

Examples

- *My boss told me to talk to talk to you.*
- *I want you to give me some advice on how to manage and organize our databases.*
- *I need you to listen to me very carefully.*
- *Is there anything else you can advise us to do?*

The Hash Function

The **hash function** creates a value also known as the hashing algorithm which is very useful when searching databases.

A **hash function** normally groups a number of characters, known as a key, and relates them to a value of a determined length, called a hash value or hash. This hash value represents the original string of characters or key, however, it is usually shorter than the original. The hash function is also referred to as hashing.

Hashing is used for tracing elements and creating indexes in databases since it is simpler to find the abbreviated version of the item or hash value, than the longer original value. In other words, hashing is used with a database to have items retrieved more quickly, by using a shorter version of such item.

Example

- *In addition, you can also use the hashing function; it is helpful to have items retrieved faster, by using a shorter version of such item.*

